

GANDAKI COLLEGE OF ENGINEERING AND SCIENCE

LABORATORY REPORT



AGILE SOFTWARE DEVELOPMENT LAB

Lab No. 4

BESE III YEAR, VI-SEM

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Submitted To:

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NAME OF EXPERIMENT: Implementation of Agile Project Management Tools (JIRA, Asana, Trello, ClickUp, Zoho Projects, etc.)

AIM: To study and implement Agile Project Management using tools such as **Jira, Asana, Trello, ClickUp, and Zoho Projects** for planning, organizing, assigning, tracking, and managing software development tasks effectively.

Objectives:

1. To understand the concept of Agile Project Management.
2. To study the features of popular Agile Project Management tools.
3. To create and manage a software project using an Agile tool.
4. To create tasks, assign responsibilities, and set deadlines.
5. To organize project workflows using boards and task lists.
6. To monitor project progress using Agile methodologies.
7. To improve collaboration and communication among team members.

THEORY:

Introduction to Agile Project Management

Agile Project Management is an iterative approach to planning, developing, and managing projects. It focuses on continuous improvement, customer feedback, collaboration, and delivering small, functional increments of a project. Agile helps teams adapt quickly to changing requirements while maintaining high productivity and software quality.

Agile project management tools simplify task planning, sprint management, progress tracking, and communication among team members.

1. Jira

Jira is one of the most widely used Agile project management tools developed by Atlassian. It supports Scrum and Kanban methodologies and is commonly used for issue tracking, sprint planning, and bug management.

Features:

- Sprint planning
- Backlog management
- Kanban and Scrum boards
- Bug and issue tracking
- Reports and dashboards

2. Asana

Asana is a cloud-based project management tool used to organize and track tasks. It enables teams to assign work, set deadlines, monitor progress, and collaborate efficiently.

Features:

- Task assignment
- Project timelines
- Calendar view
- Team collaboration
- Progress tracking

3. Trello

Trello is a visual project management tool based on the **Kanban** methodology. Projects are organized into **Boards**, **Lists**, and **Cards**, making task management simple and intuitive.

Features:

- Kanban boards
- Drag-and-drop task management
- Labels and checklists
- Due dates
- Team collaboration

4. ClickUp

ClickUp is an all-in-one productivity and project management platform that combines task management, documentation, goal tracking, and reporting.

Features:

- Task and project management
- Time tracking
- Goal management
- Multiple project views
- Document management

5. Zoho Projects

Zoho Projects is an online project management application that helps teams plan projects, assign tasks, monitor progress, and collaborate effectively.

Features:

- Task scheduling
- Gantt charts
- Issue tracking
- Document sharing
- Team collaboration

Advantages of Agile Project Management Tools

- Improves team collaboration.
- Enhances task organization.
- Tracks project progress in real time.
- Simplifies sprint and backlog management.
- Increases productivity and transparency.
- Helps meet project deadlines efficiently.

Algorithm

1. Create an account on an Agile project management tool (e.g., Trello or Jira).
2. Create a new project or workspace.
3. Define the project name and description.
4. Create task lists or project boards (e.g., To Do, In Progress, Done).
5. Add project tasks or user stories.
6. Assign tasks to team members.
7. Set priorities and deadlines.
8. Update task status as work progresses.
9. Monitor project progress using dashboards or boards.
10. Complete all tasks and review project status.
11. End.

CODE:

Implementation (Example Using Trello)

Step 1: Create a Board

Board Name:

Software Engineering Lab Project

Step 2: Create Lists

To Do

In Progress

Testing

Completed

Step 3: Add Task Cards

Card	Description
Create Login Page	Design login interface
Develop Database	Create database schema
API Integration	Connect frontend and backend
Testing	Perform software testing
Documentation	Prepare project report

Step 4: Assign Members

Member A → Login Module

Member B → Database

Member C → Testing

Member D → Documentation

Step 5: Move Tasks

To Do



In Progress



Testing



Completed

Sample Workflow

Project Created Successfully



Tasks Added



Members Assigned



Progress Updated



Project Completed

Sample Output

Project Name:

Software Engineering Lab Project

Board Created Successfully

Lists Created:

✓ *To Do*

✓ *In Progress*

✓ *Testing*

✓ *Completed*

Tasks Assigned Successfully

Project Progress Updated

Project Completed Successfully

RESULT

The implementation of Agile Project Management using tools such as Trello, Jira, Asana, ClickUp, or Zoho Projects was successfully carried out. Tasks were created, assigned, monitored, and managed using Agile workflows, demonstrating effective project planning and team collaboration.

CONCLUSION

This experiment provided practical experience in using Agile Project Management tools for organizing and managing software development projects. By creating project boards, assigning tasks, tracking progress, and monitoring project completion, the experiment demonstrated how tools such as Jira, Asana, Trello, ClickUp, and Zoho Projects support Agile methodologies. These tools improve collaboration, enhance project visibility, simplify workflow management, and contribute to the timely delivery of high-quality software projects.